COURSE TITLE : QUALITY MANAGEMENT & STANDARDIZATION
COURSE CODE : 6102
COURSE CATEGORY : A
PERIODS/ WEEK : 6
PERIODS/ SEMESTER : 90
CREDIT : 6

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Quality and standardization	20
2	Quality standards	25
3	Material standardization	25
4	Total process standardization	20
TOTAL		90

Course Outcome :

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To comprehend standardization and quality
2	To analyze quality standards
3	To comprehend materials and their impact on quality
4	To apply quality standards

MODULE I QUALITY AND STANDARDIZATION

1.1.0 To comprehend standardization and quality

1.1.1. To define quality
1.1.2. To define standardization
1.1.3. To explain need of standardization
1.1.4. To describe quality assessment methods
1.1.5. To identify customer satisfaction
1.1.6. To list various quality and process research organizations and firms, in printing and allied fields
1.1.7. To list various quality standards
1.1.8. To explain different quality standards in printing industry

MODULE II QUALITY STANDARDS

2.1.0 To analyze quality standards

2.1.1 To describe different process quality parameters
2.1.2 To identify densitometry and spectrophotometry
2.1.3 To explain different print properties and their measurement
2.1.4 To explain different quality control patches and targets

MODULE III MATERIAL STANDARDIZATION

3.1.0 To comprehend materials and their impact on quality

- 3.1.1 To define importance of material standardization
- 3.1.2 To explain different properties of paper and their measurement
- 3.1.3 To explain different properties of ink and their measurement
- 3.1.4 To explain different properties of plastics and their measurement

MODULE IV   TOTAL PROCESS STANDARDIZATION

4.1.0 To apply quality standards

- 4.1.1 To define the quality and process standardization methods and techniques
- 4.1.2 To explain Six Sigma methodology
- 4.1.3 To explain Lean printing
- 4.1.4 To describe the application of lean and Six sigma in print production
- 4.1.5 To explain green printing
- 4.1.6 To describe how to reduce wastes during printing
- 4.1.7 To explain color management with printing press

CONTENT DETAILS

MODULE   I

Quality : ISO definition. Standardization – ISO definition. Need of quality control. Advantages of quality control/standardization. Quality assessment methods – subjective (visual), objective, measured.

How a customer determine the quality. Customer satisfaction.

Factors of influence and specifications determining the quality of the offset print.

Introduction to Different process/quality/standard/material research firms/organization – ISO, GATF, FOGRA, UGRA, PIRA, Brunner, IGT, CIE, ICC, Printing Industries of America (PIA).

Industry standards for quality – introduction to ISO (12647-series, 2846-series, 13655, 15930 – series, 12640-series, 12642 & 166614 – introduction only, 19306, 16621 & 12636, 16620 & 12218).

MODULE   II

Different print properties: Density – definition, formula. Dot gain – definition, types, formula and calculation. Dot area/area coverage/tone value – definition, formula (Murray-Davies formula). Yule-Nielsen equation for plate area coverage. Ink trapping – definition, formula. Print contrast – definition, formula. Grey balance and Hue error – definition, reason of hue error, formula. Ghosting – definition, formula (Ghosting factor). Ink film thickness and density.

<u>Densitometry and spectrophotometry:</u> Densitometry definition. Optical Density. Application and need of densitometry. Densitometers – types of densitometers, usage and functions of modern densitometers. Polarizing filter and its use in densitometry. Densitometer manufacturing companies. Spectrophotometry – definition & need. Color measurement. Spectrophotometers. Functions of modern spectrophotometers. Use of densitometer and spectrophotometer for quality evaluation. Color difference - Delta E.

<u>Quality & process control targets / patches:</u> (only introduction and use of) Printing Industries RHEM Light Indicator. Printing Industries QC Photos. Printing Industries Color Communicator. HRR Pseudoisochromatic Test (color blindness). Munsell Color Checkers – photography, graphic art, digital imaging. Printing Industries Proof Comparator. GATF/Printing Industries Plate Control Target. UGRA/FOGRA media wedge. UGRA plate control target film. UGRA/FOGRA process control elements in patches – (information panel, type patch, resolution panel, line panels, checkered panels, doubling/Slur patches, fine dot patches, continuous tone wedge, halftone wedge, visual reference steps, color control patches, register patch). UGRA/FOGRA Digital plate control wedge. UGRA/FOGRA digital print scale. Color control Bars – parts of color control bar – tint patches, solid patches, grey patches, star targets, RGB overprints, total area coverage patches. GATF CTP printing plate exposure wedge. GATF Grey balance patch. GATF ICC color charts. GATF test forms - contains different targets and sizes. GATF star target. GATF line screen (visual and measured). Brunner zebra strip. Brunner color control strip. Brunner CTP plate control strip. Brunner CMYK marks. Registration patches for unit to unit registration. Fogra dampening control test for me.

MODULE III

The need of material standardization. Role of material properties in print quality.

Paper properties and testing methods:

Physical properties : Grammage – definition, testing equipment and method. Thickness – definition, importance, testing equipment and method. Bulk – definition, importance, testing equipment and method. Wire side and felt side. Grain direction and cross direction – definition, testing methods, importance. Curl – definition, importance, testing method. Formation. Friction – definition, importance, testing equipment and method. Dimensional stability. Compressibility – definition, importance, testing equipment and method. Smoothness – definition, importance, testing equipment and method. Porosity. Absorbency – definition, importance, testing equipment and method.

a) Strength properties : Stiffness – definition, importance, testing equipment and method. Tensile strength – definition, importance, testing equipment and method. Bursting strength – definition, importance, testing equipment and method. Tear resistance – definition, importance, testing equipment and method. Folding endurance – definition, importance, testing equipment and method. Surface strength – definition, importance, testing equipment and method.
b) Optical properties : Gloss – definition, importance, testing equipment and method. Brightness – definition, importance, testing equipment and method. Whiteness – definition, importance,

testing equipment and method. Color – definition, importance, testing equipment and method. Fluorescence. Opacity – definition, importance, testing equipment and method.

c) Chemical properties: PH of paper – definition, importance, testing equipment and method. Ash content. Moisture content – definition, importance, testing equipment and method. Permanence of paper.
d) Printing properties: Runnability. Printability.

Ink properties and testing methods:

a) Drying properties: Absorption, solvent evaporation, Oxidation and Polymerization, Precipitation, Cold Setting/Hot melt, Radiation curing (UV, IR).
b) Rheology: Viscosity – definition, importance, testing equipment and methods (viscometers). Yield value. Thixotropy. Tack– definition, importance, testing equipment and methods (tack meter/inkometer.
c) Color of ink.
d) End user properties: Rub resistance– definition, importance, testing equipment and methods. Ink film thickness – definition, importance, testing equipment and methods. Printability. Light fastness.

Types of Plastics and properties: PVC, PE, LDPE, LLDPE, HDPE, PP, PET, Polystyrene.

MODULE IV

<u>Quality/process standardization methods & techniques:</u> Optimization. Lean printing – Lean manufacturing concept, definition, Advantages. Lean printing definition. Lean goals. 7 types of wastes in Lean. Lean tools – 5s, standardized work instructions, value stream mapping, Total proactive maintenance, Kaizen events, Just in time management, 5 whys, etc. Six Sigma – Six Sigma concept, definition, advantages. Six Sigma team. DMAIC & DMADV process (in detail). Practical case study of Lean printing and Six Sigma. Total Quality management (TQM). Application of Lean, Six sigma, TQM etc.

<u>Green printing:</u> Use of echo friendly paper. Use of echo friendly inks. Use of echo friendly solvents. IPA substitution. Available echo friendly - paper, ink, solvents, IPA substitutes.

<u>Waste Reduction:</u> Learn to save paper. Importance and advantage of saving paper. Waste reduction measures in – Prepress, Press, Post-press. Use of Lean manufacturing to reduce waste.

<u>Color management & Digital QC:</u> Color management concept, need and importance. Calibrating printing machine (offset). 8 steps of press color management.

Reference :		
Author	**Title**	**Publishers**
R. H. Leach	The Printing Ink Mannual 5th Edition	Chapman &Hall, London
Fonaid E Tood	Printing inks	PIRA international united kingdom
Bob thomson	Printing Material Science and Technology	PIRA
W.H. Bureau	What the printer should know about the paper	GATF
Michael Bernard, John Peacock.	Handbook of printing and production	
Helmut Kipphan	Hand book of print media	Springer Science & Business Media
Heigh. M. Speir	Introduction in Printing Technology	
Adams J.M,	Printing Technology 5th edition	Delmar Publishers, NewYork
Kevin kooper	Lean printing Path way to success	PIA GATF Press
Thomas Pyzdek and Paul Keller	The Six Sigma Handbook, Third Edition	Mc Graw Hill
T.M. Kubiak	The Certified Six Sigma Black Belt Handbook	Pearson second edition
Roderick A. Munro	The Certified Six Sigma Green Belt Handbook	Pearson second edition
HL Apfelberg and MJ Apfelberg	Implementing quality management in graphic arts	GATF
Prameela J Groff	Test images for printing	
ISO/TC 130	12647 – 1 to 5 standards	